

Supplementary material document for “Extended-support beta regression for $[0, 1]$ responses”

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S1 Introduction

The supplementary material for “Extended-support beta regression for $[0, 1]$ responses” is available at <https://github.com/ikosmidis/XBX-supplementary> and provides the current supplementary material document, and instructions and scripts to reproduce all numerical results and figures in the main text and the supplementary material document.

S2 XBX versus doubly-censored normal distribution

In this section, we carry out a comparison of the XBX distribution to the doubly-censored normal distribution. For each combination of $\mu \in \{0.5, 0.7, 0.9\}$, $\phi \in \{0.5, 2, 5, 20\}$, and $\nu \in \{0.01, 0.1, 1\}$, we use numerical integration to compute the expected value μ' and standard deviation σ' of the $\text{XBX}(\mu, \phi, \nu)$ distribution. Then, we numerically solve the system of equations

$$\begin{aligned} E(Y' \mid m, s) &= \mu' \\ \sqrt{\text{var}(Y' \mid m, s)} &= \sigma' \end{aligned}$$

with respect m and s , where Y' has a doubly-censored normal distribution in $(0, 1)$ with mean m and standard deviation s for the uncensored Normal distribution. In this way, for each combination of μ , ϕ and ν , we can compare XBX and doubly-censored normal distributions that have the same mean and variance. The outcome of the comparison is shown in Figure S1 (densities) and Figure S2 (histograms of samples of size 1000), which can be reproduced using the script `xbx-vs-cn.R` in the supplementary material.

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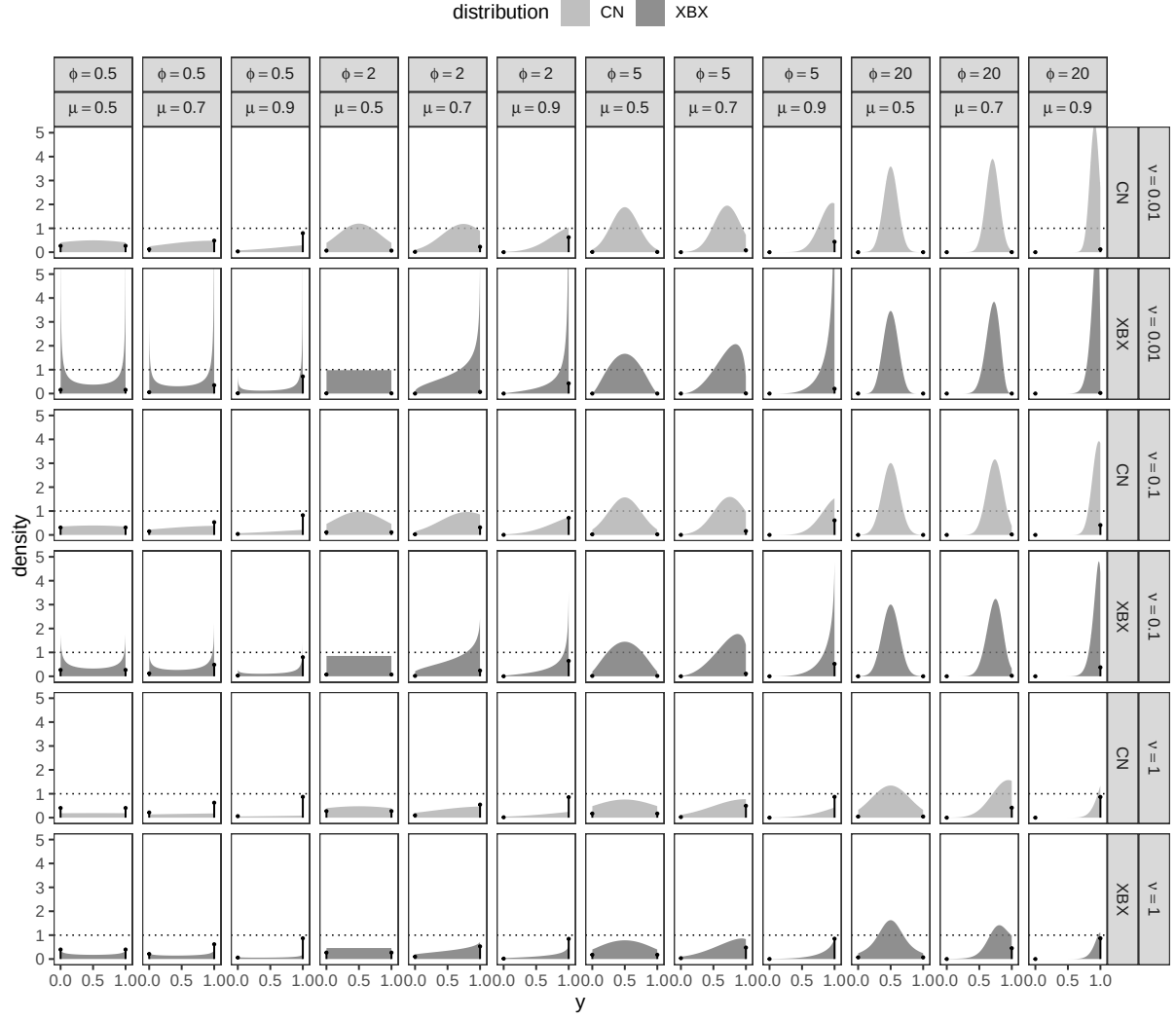


Figure S1: Densities of the XBX distribution (dark grey) and of the doubly-censored normal distribution (light grey) for all combinations of $\mu \in \{0.5, 0.7, 0.9\}$, $\phi \in \{0.5, 2, 5, 20\}$, and $\nu \in \{0.01, 0.1, 1\}$. The mean and the variance of the doubly-censored normal distribution are numerically matched to be equal to those of the XBX distribution; see Section S2 for details. The horizontal dotted line is at 1.

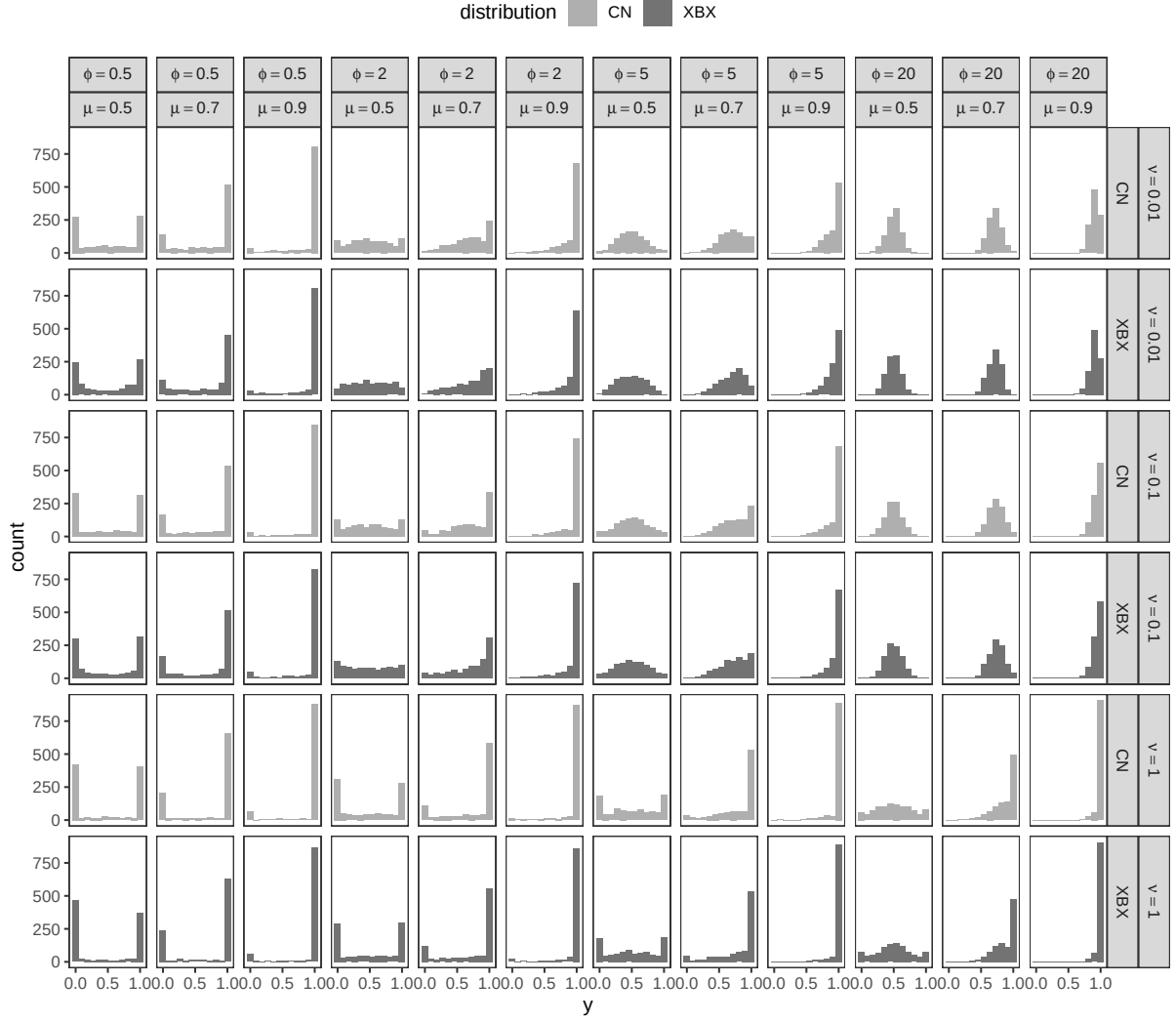


Figure S2: Histograms of samples of size 1000 from the XBX distribution (dark grey) and from the doubly-censored normal distribution (light grey) for all combinations of $\mu \in \{0.5, 0.7, 0.9\}$, $\phi \in \{0.5, 2, 5, 20\}$, and $\nu \in \{0.01, 0.1, 1\}$. The mean and the variance of the doubly-censored normal distribution are numerically matched to be equal to those of the XBX distribution; see Section S2 for details.

S3 Mean and variance of $\mathbf{XBX}$ distribution as functions of μ , ϕ , and ν

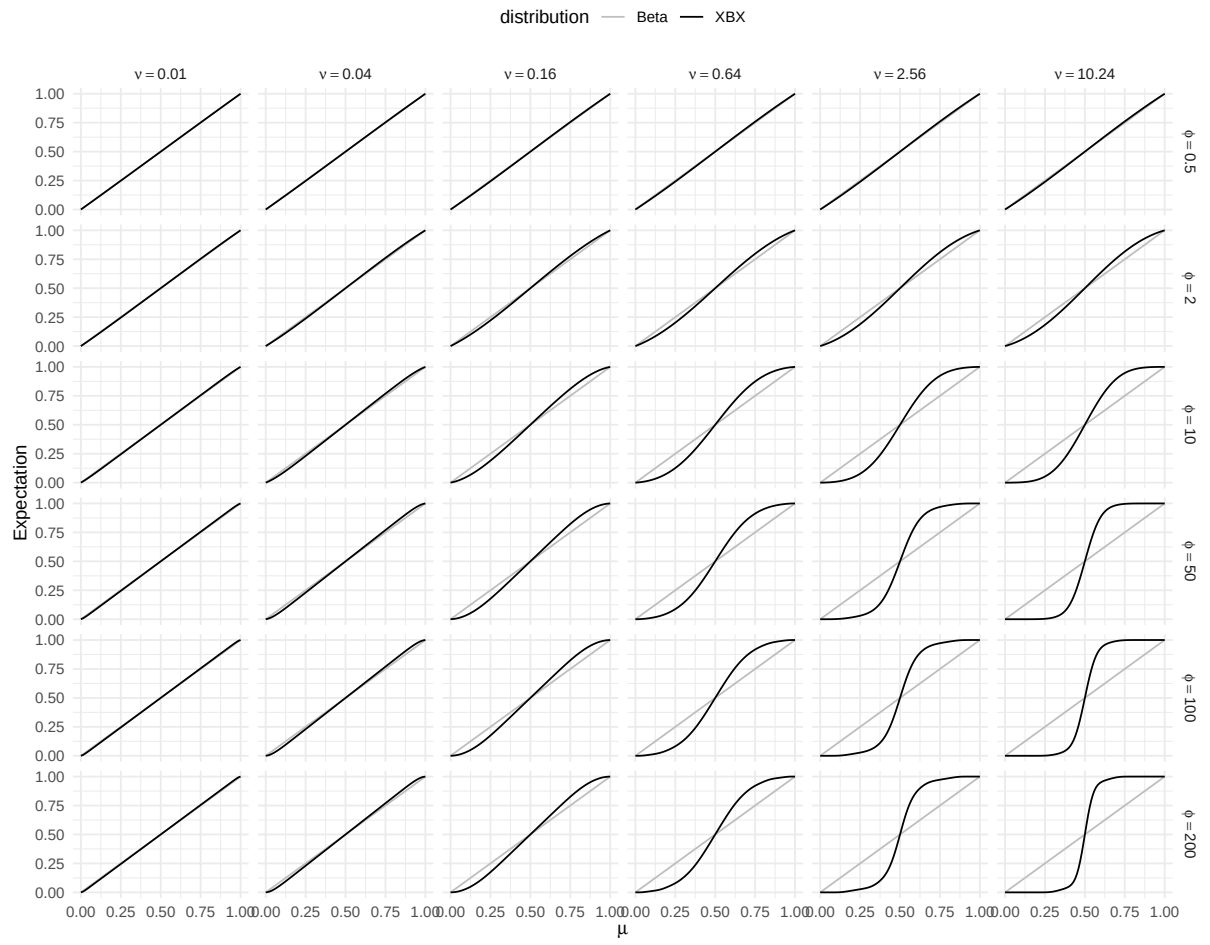


Figure S3: The mean of the XBX distribution as a function of μ for a range of values for ϕ and ν . For reference, the expectation (μ) of the beta distribution is also shown.

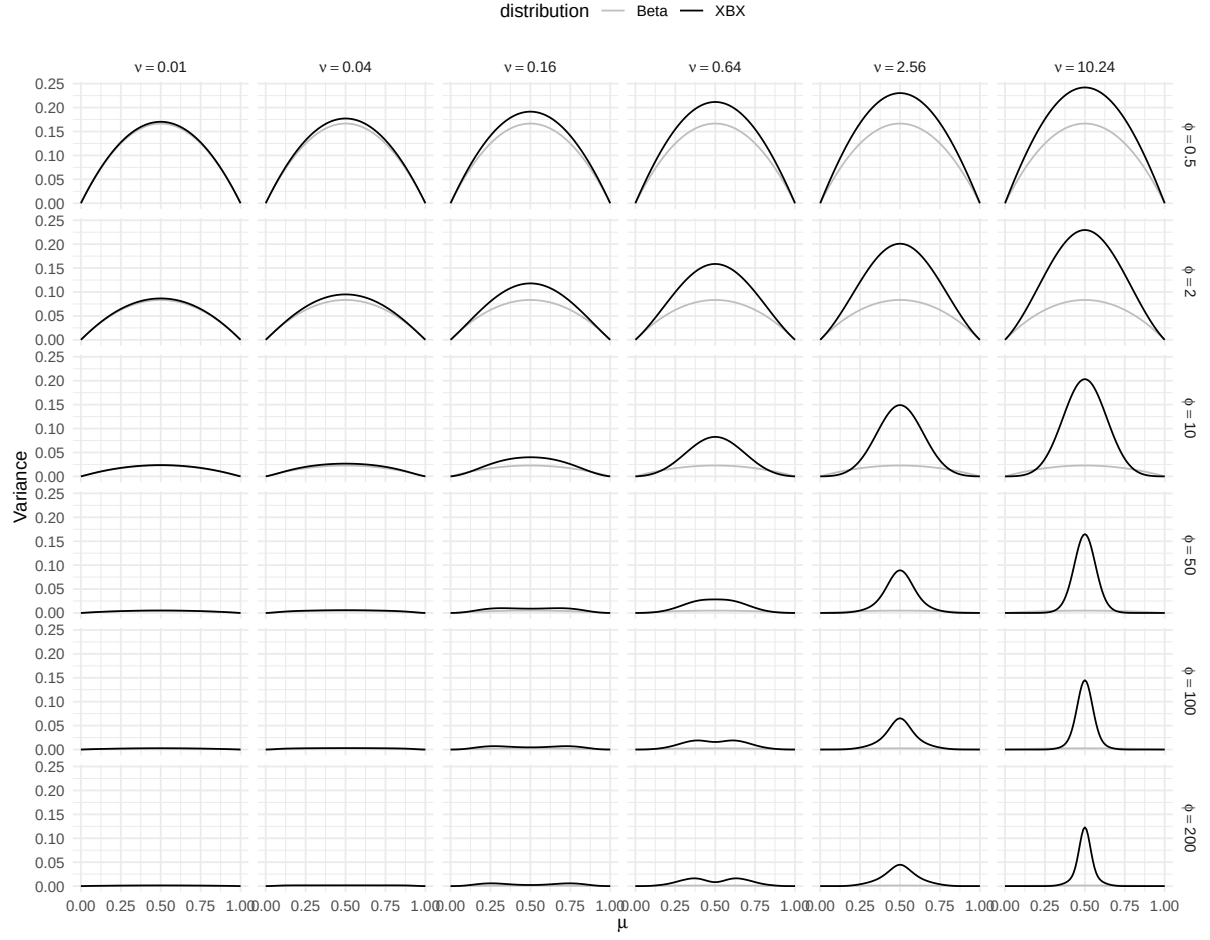


Figure S4: The variance of the XBX distribution as a function of μ for a range of values for ϕ and ν . For reference, the variance $(\mu(1 - \mu)/\phi)$ of the beta distribution is also shown as a function of μ for a range of values of ϕ .

S4 Supplementary tables and figures for Section 4

In this section, we provide supplementary plots and the full results of the loss aversion analysis in Section 4 of the main text. Specifically, Figure S5 depicts the fitted cumulative distribution functions for every model of Section 4.3, for each level of arrangement, for subjects that are male or teams with at least one male, in grade 10–12 and 16 years of age. The fitted cumulative distributions are contrasted to the corresponding empirical cumulative distribution functions obtained from the subsample of subjects that are male or teams with at least one male, in grades 10–12 and between 15 and 17 years of age. Table S2 and Table S3 show the estimates for the three-part hurdle model of Section 4.7 with a multinomial logistic regression model for the probabilities of boundary and non-boundary observations with the same covariates as those in the linear predictor (16) and predictor (17), respectively, and model B of Section 4.3 for the non-boundary observations.

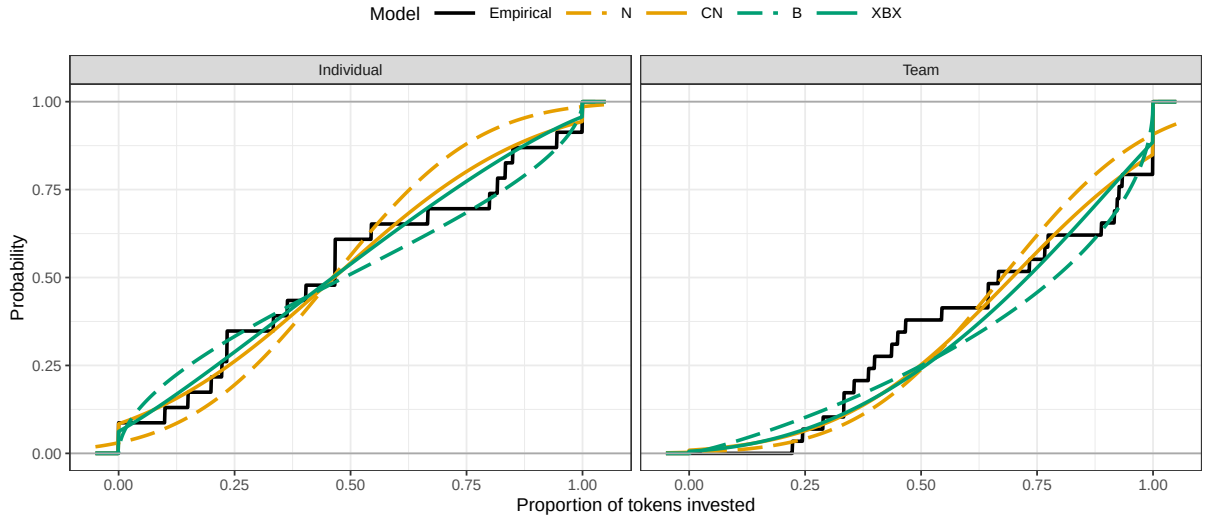


Figure S5: Fitted cumulative distribution functions for every model of Section 4.3 (N, CN, B, XBX), for each level of arrangement, for subjects that are male or teams with at least one male, in grade 10–12 and 16 years of age. The fitted cumulative distributions are contrasted to the corresponding empirical cumulative distribution functions (black, dotted) obtained from the subsample of subjects that are male or teams with at least one male, in grades 10–12 and between 15 and 17 years of age.

Table S1: Estimated arrangement effects for subjects that are male or teams with at least one male, in grade 10–12 and 16 years of age. The rows for $P(Y < 0.05)$ show model-based estimates of the probability to invest less than 5% of the tokens. All effects are contrasted to the corresponding empirical quantities obtained from the subsample of subjects that are male or teams with at least one male, in grades 10–12, and between 15 and 17 years of age. The rows for the parameters show the estimates of the distributional parameters for each arrangement setting.

	Arrangement	Empirical	N	CN		B	XBX	
$P(Y < 0.05)$	Individual	0.087	0.047	0.109		0.135	0.102	
	Team	0	0.006	0.014		0.015	0.011	
Parameters	Individual		μ	0.461	0.465	μ	0.492	0.475
			σ	0.246	0.337	ϕ	1.091	2.602
						ν		0.103
	Team		μ	0.675	0.696	μ	0.700	0.662
			σ	0.246	0.294	ϕ	1.637	3.602
						ν		0.103

Table S2: Coefficient estimates, estimated standard errors (in parentheses), maximized log-likelihood, AIC, and BIC for the three-part hurdle, with a multinomial logistic regression model for the probabilities of boundary and non-boundary observations with the same covariates as those in the linear predictor (16), and the beta regression model B for the non-boundary observations; see Section 4.7. Estimates for the multinomial regression part have been obtained using the VGAM R package Yee (2010); the standard errors that are greater than 1000 are in reality infinite due to separation, as are the corresponding estimates.

	Zero-and-one-inflated beta			
	$\log(\pi_0/\pi_{0,1})$	$\log(\pi_1/\pi_{0,1})$	$\text{logit}(\mu)$	$\log(\phi)$
(Intercept)	−22.192 (14 749.887)	−23.232 (10.334)	−0.858 (0.597)	1.209 (0.087)
G_i (Grade)	22.265 (14 749.888)	11.526 (10.841)	−2.195 (1.102)	0.222 (0.115)
T_i (Arrangement)	−0.379 (2642.760)	−0.744 (0.960)	0.278 (0.112)	0.422 (0.131)
A_i (Age)	0.060 (1158.335)	1.361 (0.753)	0.045 (0.047)	
S_i (Sex)	0.759 (0.759)	1.885 (0.549)	0.318 (0.089)	−0.292 (0.122)
$G_i T_i$ (Grade $\times$ Arrangement)	−1.057 (2642.760)	1.539 (1.053)	0.448 (0.171)	
$G_i A_i$ (Grade $\times$ Age)	−0.264 (1158.335)	−0.885 (0.777)	0.112 (0.072)	
Log-likelihood				−46.5
AIC				143.0
BIC				251.6

Table S3: Coefficient estimates, estimated standard errors (in parentheses), maximized log-likelihood, AIC, and BIC for the three-part hurdle, with a multinomial logistic regression model for the probabilities of boundary and non-boundary observations with the same covariates as those in the linear predictor (17), and the beta regression model B for the non-boundary observations; see Section 4.7. Estimates for the multinomial regression part have been obtained using the VGAM R package Yee (2010); the standard errors that are greater than 1000 are in reality infinite due to separation, as are the corresponding estimates.

	Zero-and-one-inflated beta			
	$\log(\pi_0/\pi_{0,1})$	$\log(\pi_1/\pi_{0,1})$	$\text{logit}(\mu)$	$\log(\phi)$
(Intercept)	-21.156 (1225.251)	-5.671 (0.634)	-0.858 (0.597)	1.209 (0.087)
G_i (Grade)	17.840 (1225.251)	2.263 (0.506)	-2.195 (1.102)	0.222 (0.115)
T_i (Arrangement)	-1.430 (1.130)	0.425 (0.428)	0.278 (0.112)	0.422 (0.131)
A_i (Age)			0.045 (0.047)	
S_i (Sex)	0.919 (0.758)	1.880 (0.541)	0.318 (0.089)	-0.292 (0.122)
$G_i T_i$ (Grade $\times$ Arrangement)			0.448 (0.171)	
$G_i A_i$ (Grade $\times$ Age)			0.112 (0.072)	
Log-likelihood				-53.5
AIC				145.1
BIC				227.7

S5 XBX versus heteroscedastic two-limit tobit regression

In this section, we provide supplementary plots and the full results of the simulation experiment in Section 5 of the main text, for the comparison of XBX regression with heteroscedastic two-limit tobit regression model in terms of their predictive performance. Specifically, Figure S6 to Figure S9 show simulated data sets for all combinations of $u \in \{2^{-6}, 2^{-5}, \dots, 2\}$, and $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4. Figure S10 shows $S_{\text{CN}}/S_{\text{XBX}} - 1$, which is the average relative change in S in (18), as a function of u , when moving from XBX regression to the two-limit heteroscedastic tobit, based on their maximum likelihood fits, for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4.

The simulation experiment, Figure S6 to Figure S9, Figure S10, and Figure 6 in the main text can be reproduced using the script `xbx-vs-htobit-crss.R` in the supplementary material.

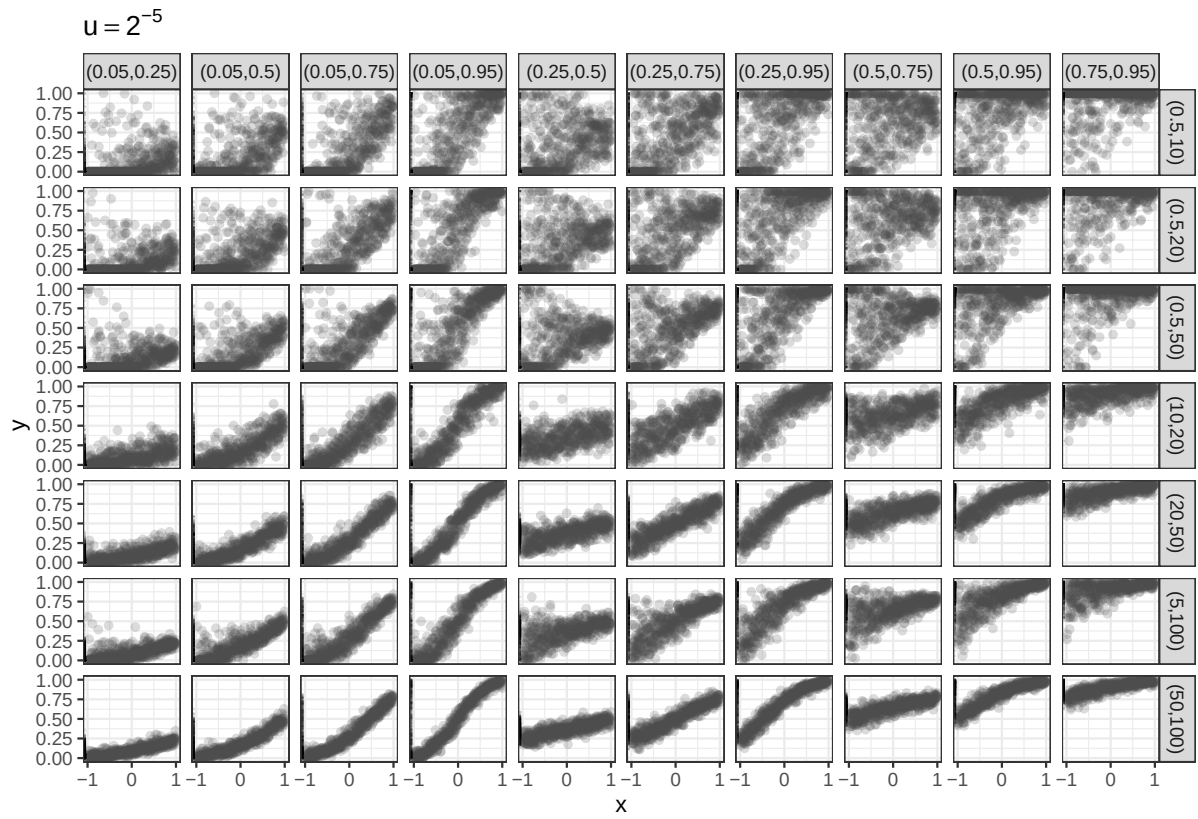
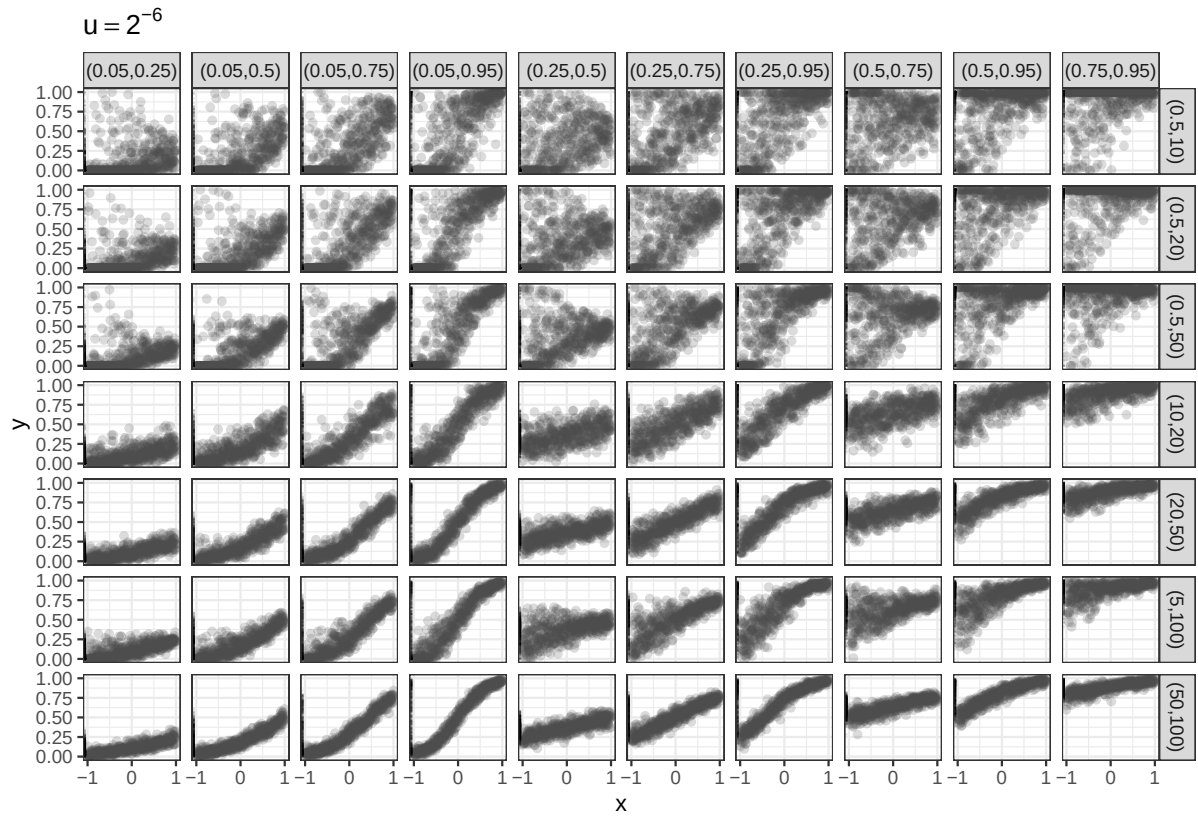


Figure S6: Scatterplots of simulated response (y) versus covariate (x) values for $u = 2^{-6}$ and $u = 2^{-5}$, for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4 of the main text.

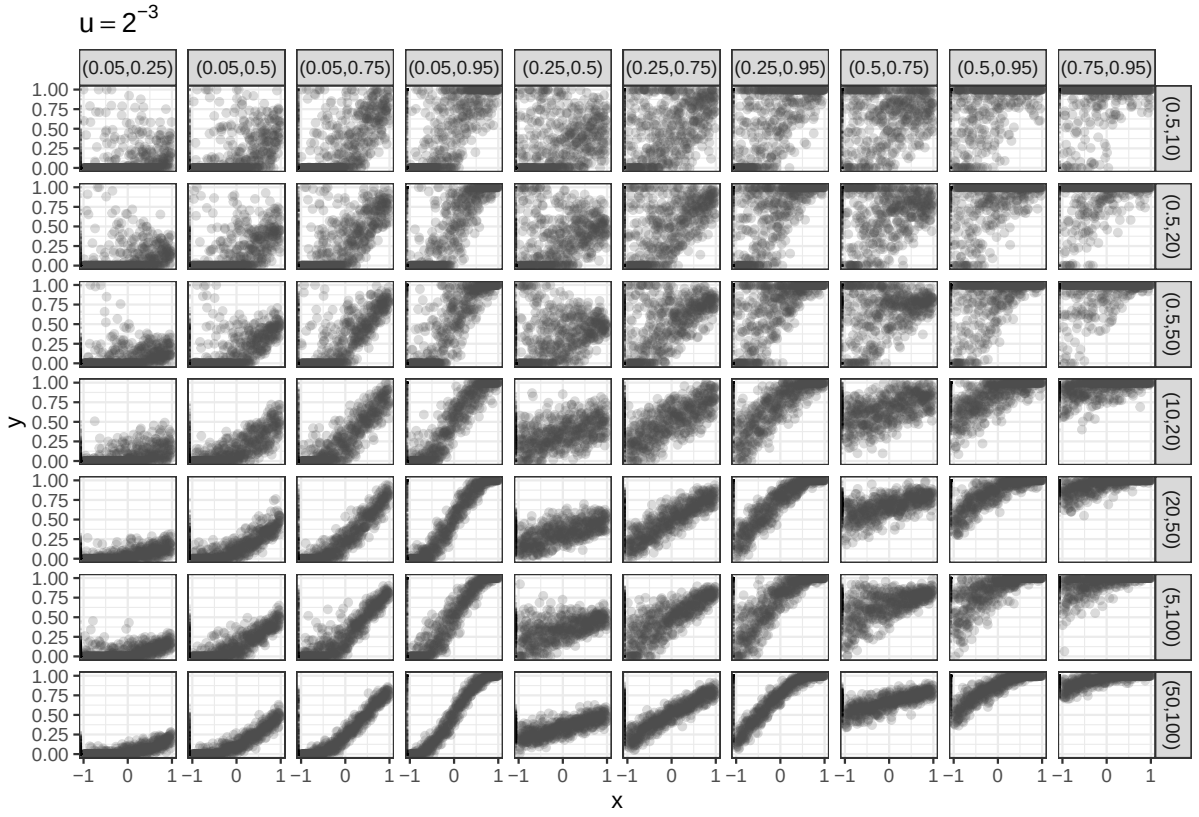
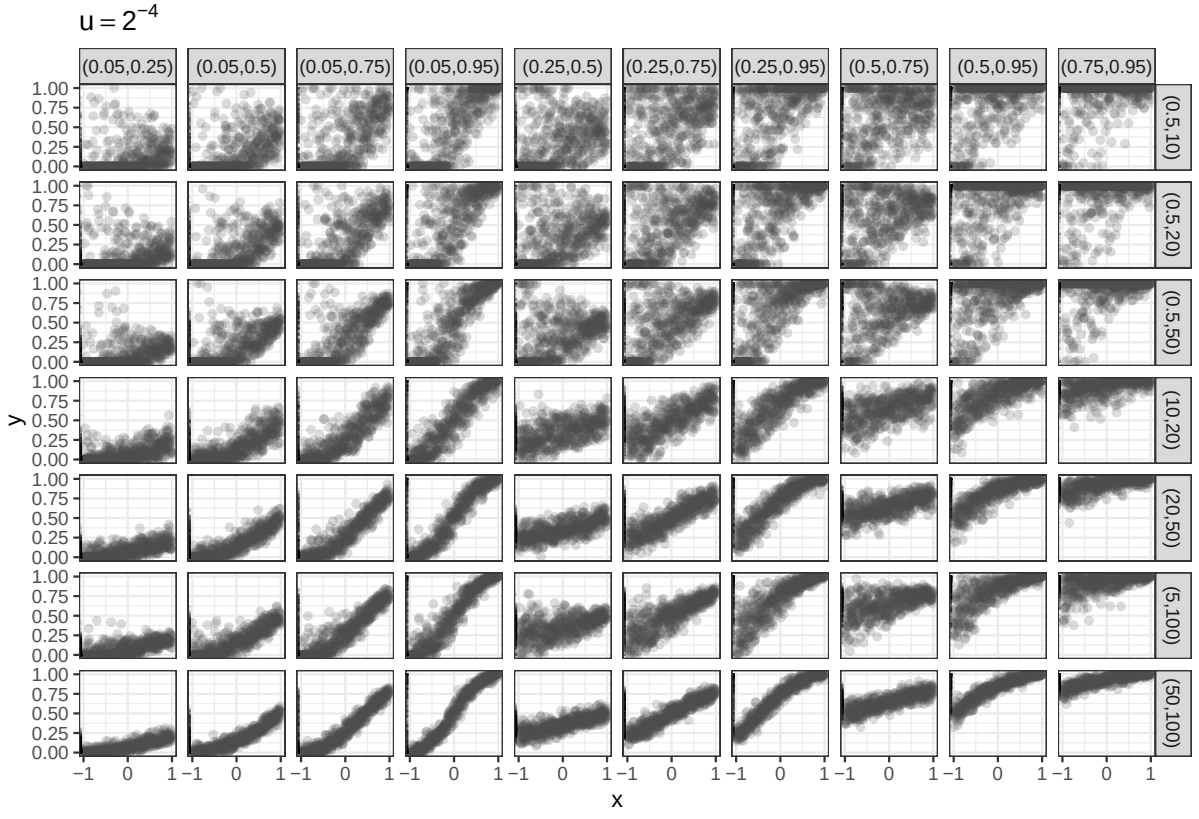


Figure S7: Scatterplots of simulated response (y) versus covariate (x) values for $u = 2^{-4}$ and $u = 2^{-3}$, for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4 of the main text.

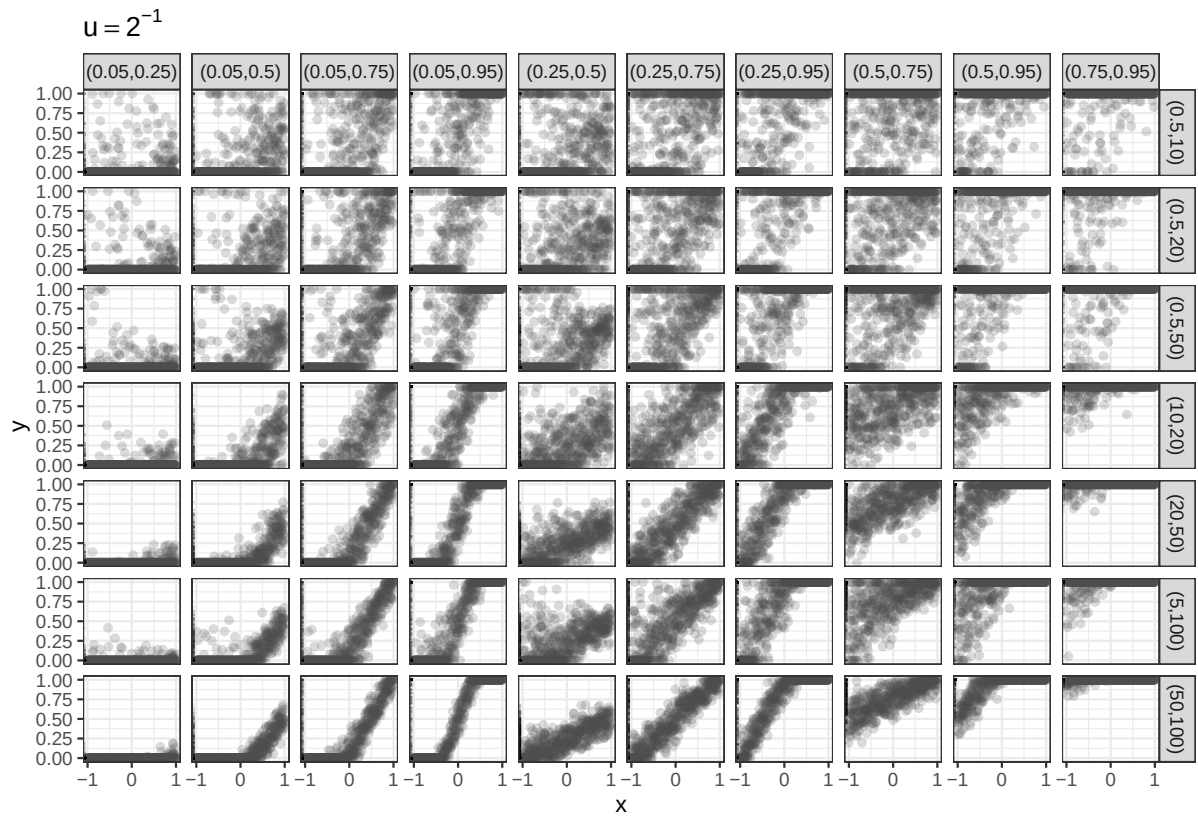
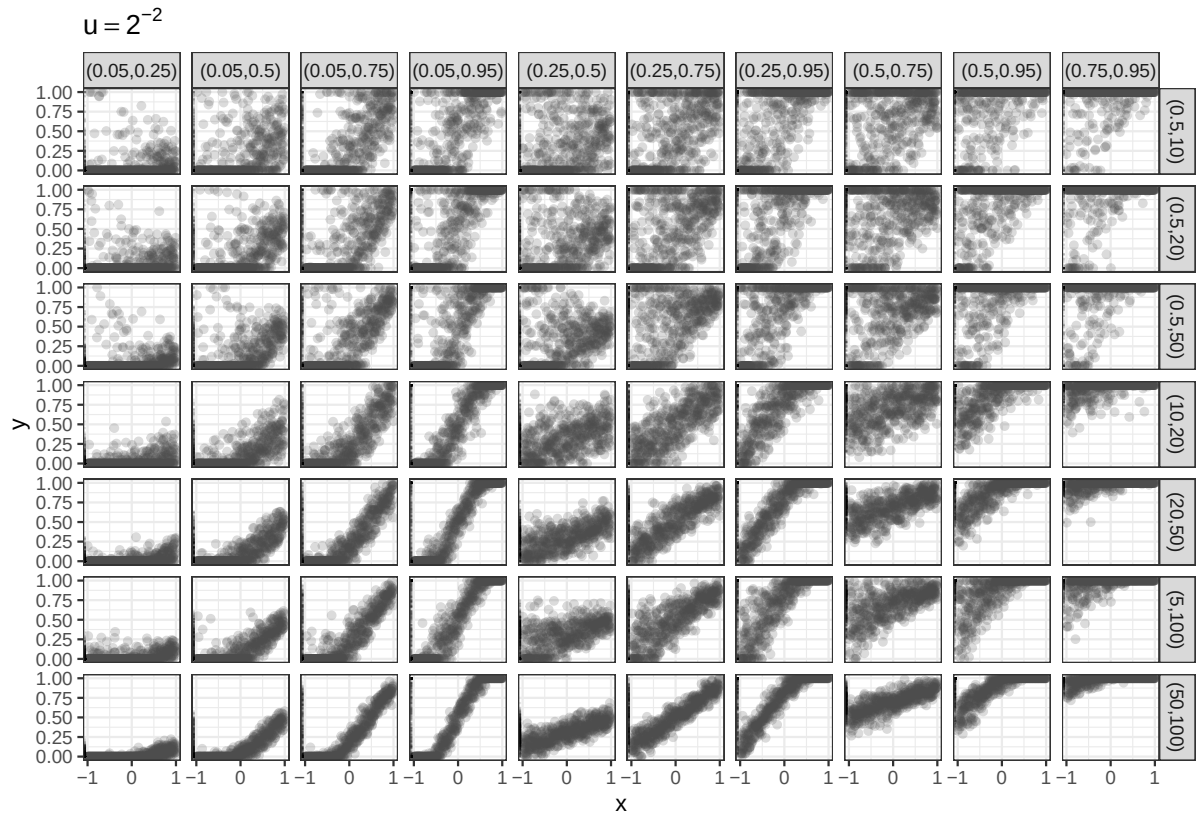


Figure S8: Scatterplots of simulated response (y) versus covariate (x) values for $u = 2^{-2}$ and $u = 2^{-1}$, for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4 of the main text.

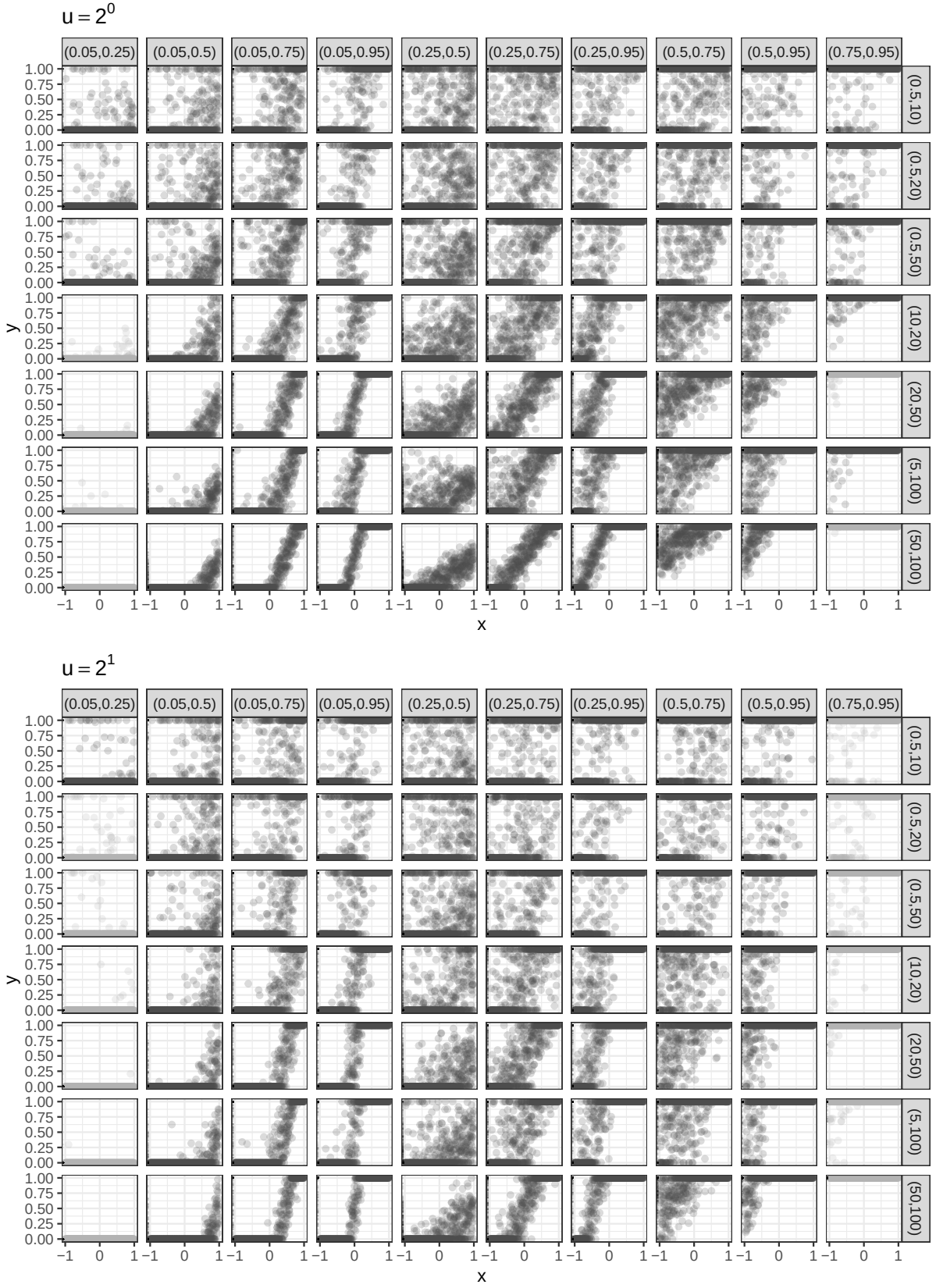


Figure S9: Scatterplots of simulated response (y) versus covariate (x) values for $u = 1$ and $u = 2$, for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4 of the main text. Light grey indicates settings where the average probability of boundary observations is larger than 0.95.

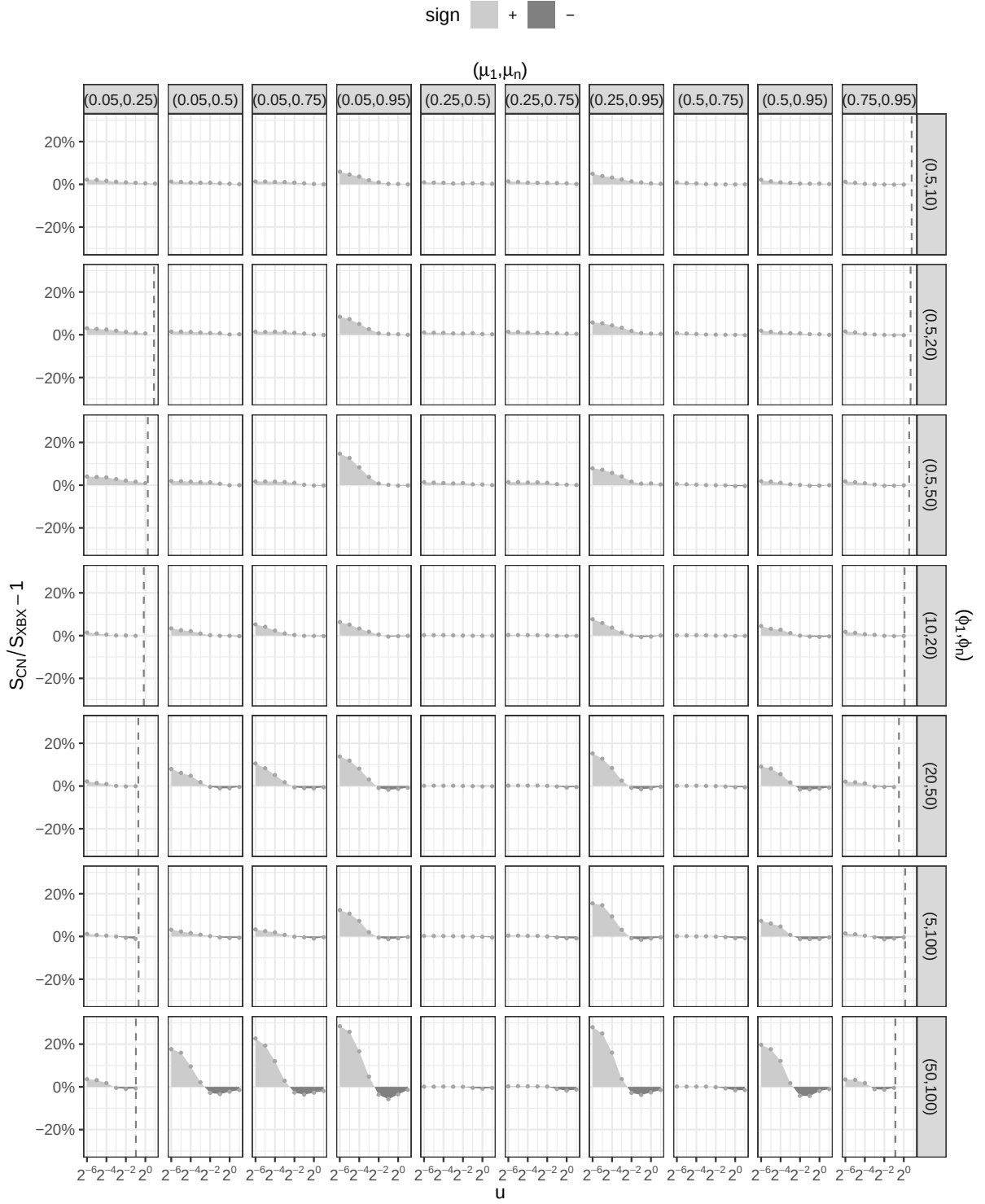


Figure S10: Average relative change in S in (18) ($S_{CN}/S_{XBX} - 1$) when moving from XBX regression to the two-limit heteroscedastic model, based on 100 samples of size $n = 500$ for each combination of $u \in \{2^{-6}, 2^{-5}, \dots, 2\}$, and $[\mu_1, \mu_n]$ (columns) and $[\phi_1, \phi_n]$ (rows) intervals in Table 4. Positive values of the relative change correspond to $S_{CN} > S_{XBX}$, and, thus, worse predictive performance of the CN model relative to the XBX regression model. The dashed vertical lines are the smallest values of u for which the average probability of boundary observations exceeds 0.95 for each combination of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$.

S6 XBX versus beta regression on rescaled responses and the normal linear regression model

In this section, we provide the results from the comparison of XBX regression to the beta regression model (B) fit on responses rescaled according to the Smithson and Verkuilen (2006) proposal, and to the normal linear regression model (N), in terms of their predictive performance using the simulation setup of Section 5. Both beta and XBX regression models use the covariates $x_1, \dots, x_n$ in the μ_i and ϕ_i regression structures. The normal linear regression model uses covariates $x_1, \dots, x_n$ in the linear predictor and constant error variance.

Figure S11 shows $S_B/S_{\text{XBX}} - 1$, and Figure S12 shows $S_N/S_{\text{XBX}} - 1$, as a function of u , for all combinations of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$ in Table 4.

Figure S11 can be reproduced using the script `xbx-vs-beta-crss.R` in the supplementary material, and Figure S12 can be reproduced using the script `xbx-vs-ols-crss.R` in the supplementary material.

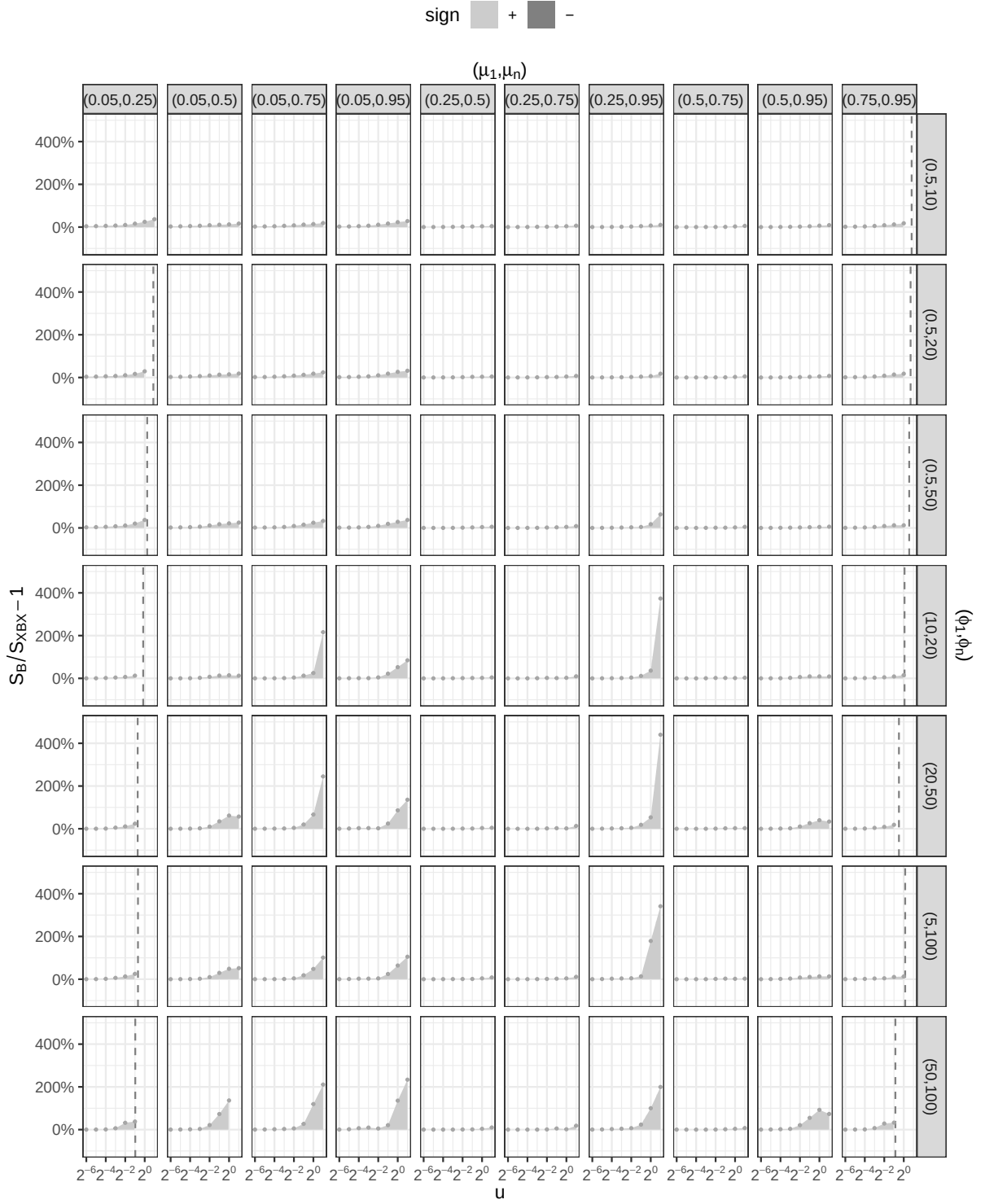


Figure S11: Average relative change in S in (18) ($S_B/S_{XBX} - 1$) when moving from the XBX regression model to the beta regression model (B) fit on rescaled responses according to the Smithson and Verkuilen (2006) proposal, based on 100 samples of size $n = 500$ for each combination of $u \in \{2^{-6}, 2^{-5}, \dots, 2\}$, and $[\mu_1, \mu_n]$ (columns) and $[\phi_1, \phi_n]$ (rows) intervals in Table 4. Positive values of the relative change correspond to $S_B > S_{XBX}$, and, thus, worse predictive performance of the B model relative to the XBX regression model. The dashed vertical lines are the smallest values of u for which the average probability of boundary observations exceeds 0.95 for each combination of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$.

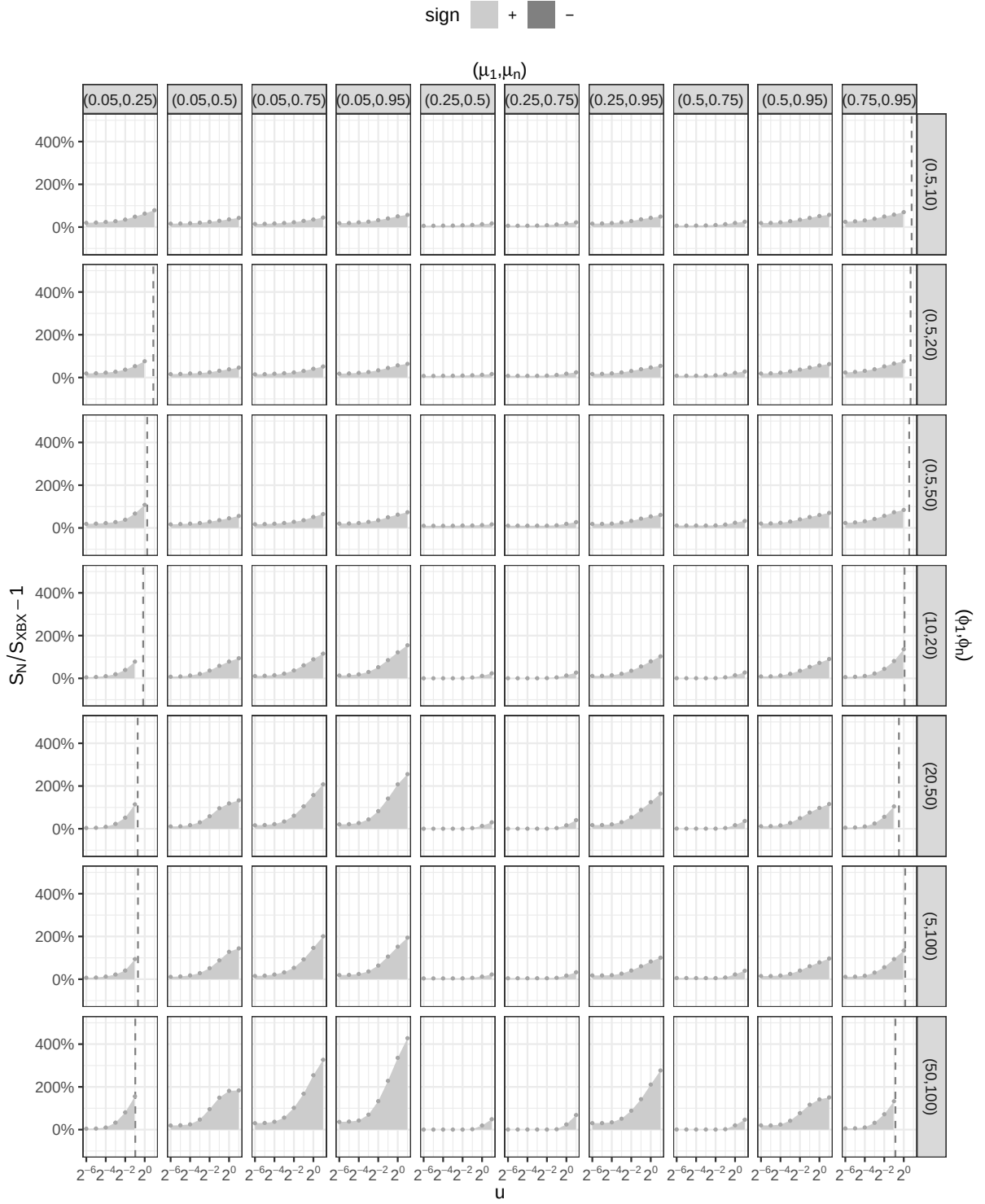


Figure S12: Average relative change in S in (18) ($S_N/S_{\text{XBX}} - 1$) when moving from the XBX regression model to the normal linear regression model (N), based on 100 samples of size $n = 500$ for each combination of $u \in \{2^{-6}, 2^{-5}, \dots, 2\}$, and $[\mu_1, \mu_n]$ (columns) and $[\phi_1, \phi_n]$ (rows) intervals in Table 4. Positive values of the relative change correspond to $S_N > S_{\text{XBX}}$, and, thus, worse predictive performance of the N model relative to the XBX regression model. The dashed vertical lines are the smallest values of u for which the average probability of boundary observations exceeds 0.95 for each combination of $[\mu_1, \mu_n]$ and $[\phi_1, \phi_n]$.

References

- Smithson M, Verkuilen J (2006). “A Better Lemon Squeezer? Maximum-Likelihood Regression with Beta-Distributed Dependent Variables.” *Psychological Methods*, **11**(1), 54–71. doi:10.1037/1082-989X.11.1.54.
- Yee TW (2010). “The VGAM Package for Categorical Data Analysis.” *Journal of Statistical Software*, **32**(10), 1–34. doi:10.18637/jss.v032.i10.